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COUNTING METHODS:

Concept and Equations	Examples
Multiplication Principle: The number of possible choices of 1 element apiece from the data sets $A_1, A_2, \cdots, A_n$ is $\lvert A_1\rvert \times \lvert A_2\rvert \times \ldots \times \lvert A_n\rvert$.	The number of possible choices of 1 element apiece from the data sets $S = \{-5, 0, 1\}$, $T = \{3, 4\}$, and $U = \{3, 4, 5, 6\}$ is $\lvert S\rvert \times \lvert T\rvert \times \lvert U\rvert = 3 \times 2 \times 4 = 24$.
Factorial: $n! = n \times (n-1) \times \ldots \times 1$ $0! = 1! = 1$ $n! = (n-1)!(n)$	4! = 4 × 3 × 2 × 1 = 24 4! = 3! × 4
Permutations: A data set of n objects has $n!$ permutations	The data set of letters A, B, and C has 3! = 6 permutations: ABC, ACB, BAC, BCA, CAB, and CBA.
Combinations: The number of possible choices of k objects from a data set of n objects is $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$.	The number of 2-element subsets of data set {A, B, C, D, E} is $\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{(2)(6)} = 10.$ The 10 subsets are: {A, B}, {A, C}, {A, D}, {A, E}, {B, C}, {B, D}, {B, E}, {C, D}, {C, E}, and {D, E}.

Probability and Sequences

PROBABILITY:

Concept	Definition, Notation, and Equations	Example: Rolling a die with 6 numbered sides once
Event	A set of outcomes of an experiment	The event of the outcome being an odd number is the set {1, 3, 5}.
Probability	The probability $P(E)$ of an event E is a number between 0 and 1, inclusive. If each outcome is equally likely, $P(E) = \frac{\text{(the number of possible outcomes in E)}}{\text{(the total number of possible outcomes)}}$.	If the 6 outcomes are equally likely, the probability of each outcome is $\frac{1}{6}$. The probability that the outcome is an odd number is $P(\{1, 3, 5\}) = \frac{\lvert\{1, 3, 5\}\rvert}{6} = \frac{3}{6} = \frac{1}{2}$.
Conditional Probability	The probability that E occurs if F occurs is $P(E\lvert F) = \frac{\lvert E \cap F\rvert}{\lvert F\rvert}$.	$P(\{1, 3, 5\}\lvert\{1, 2\}) = \frac{\lvert\{1\}\rvert}{\lvert\{1, 2\}\rvert} = \frac{1}{2}$

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Not E	The set of outcomes not in event E: $P(\text{not } E) = 1 - P(E)$.	$P(\text{not}\{3\}) = \frac{6-1}{6} = \frac{5}{6}$
E and F	The set of outcomes in both E and F, that is, $E \cap F$; $P(E \text{ and } F) = P(E \cap F) = P(E\|F)\, P(F)$.	For $E = \{1, 3, 5\}$ and $F = \{2, 3, 5\}$: $P(E \text{ and } F) = P(E \cap F) = P(\{3, 5\}) = \frac{\|\{3, 5\}\|}{6} = \frac{2}{6} = \frac{1}{3}$
E or F	The set of outcomes in E or F or both, that is, $E \cup F$; $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$.	For $E = \{1, 3, 5\}$ and $F = \{2, 3, 5\}$: $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$ $= \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}$.
Dependent and Independent Events	E is dependent on F if $P(E\|F) \neq P(E)$. E and F are independent if neither is dependent on the other. If E and F are independent, $P(E \text{ and } F) = P(E)P(F)$.	For $E = \{2, 4, 6\}$ and $F = \{5, 6\}$: $P(E\|F) = P(E) = \frac{1}{2}$, and $P(F\|E) = P(F) = \frac{1}{3}$, so E and F are independent. Thus $P(E \text{ and } F) = P(E)P(F) = \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{1}{6}$.

SEQUENCE:

An ordered list of elements, starting with a first element.

Example: The function $a(n) = n^2 + \left(\frac{n}{5}\right)$ with the domain of all positive integers $n = 1, 2, 3, \ldots$ is an infinite sequence a_n.

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